

not be able to effectively capture the temporal dynamics present in videos. These include motion, action, and scene dynamics. Sequence models can model the temporal dependencies between frames, enabling them to capture these dynamics effectively.

Use of sequence models alongside image models is vital, and their applications are found in action recognition, video captioning, or identifying specific objects or events within a video. Figure 5 gives an example of video analysis using ViT.

Contrastive language-image pretraining (CLIP)

In recent years, there has been a surge of interest in multimodal machine learning, which aims to develop models that can understand and generate information from both textual and visual inputs. CLIP has gained significant attention for its ability to bridge the gap between natural language processing (NLP) and computer vision (CV), enabling a more comprehensive understanding of multimodal data. It leverages the power of large-scale unsupervised pretraining to learn joint representations of images and their associated textual descriptions. The approach relies on a contrastive learning framework, which trains a model to pull together pairs of matching images and text while pushing apart non-matching pairs. By learning to associate relevant images and textual descriptions, CLIP can form a rich and coherent representation of the underlying semantic content much like the ‘zero-shot’ capabilities of GPT-2 and GPT-3.

Simple contrastive learning (SimCLR)

Self-supervised learning has emerged as a powerful paradigm in the field of deep learning, enabling models to learn meaningful representations from unlabelled data. SimCLR is one such self-supervised learning model.

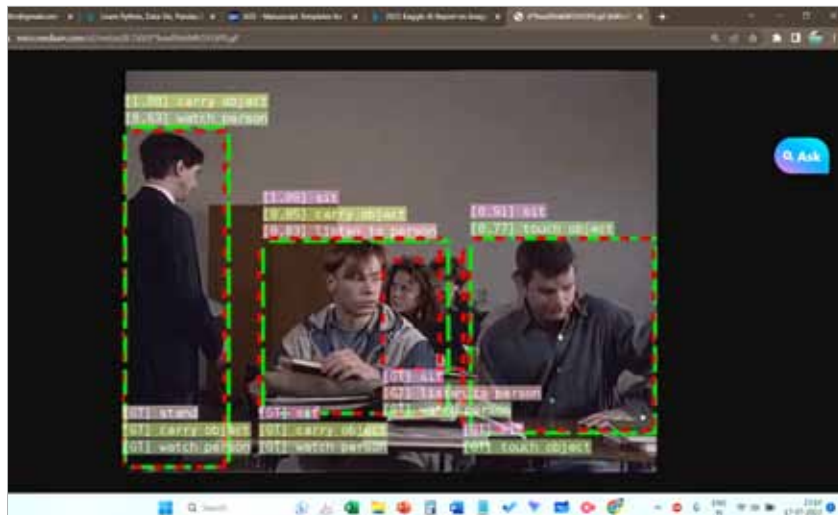


Figure 5: Video analysis using ViT

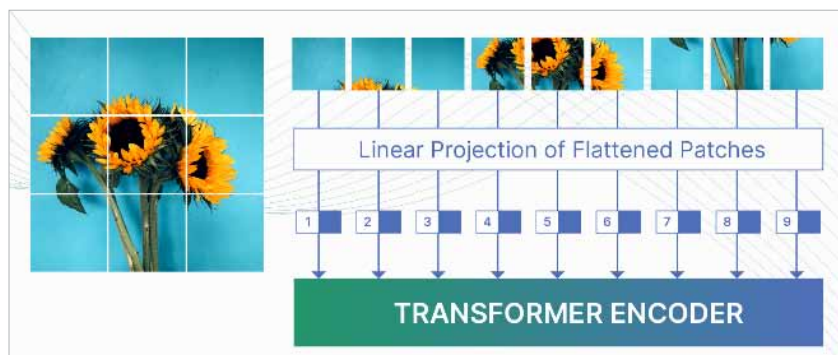


Figure 6: Vision transformer (ViT)

It has gained a significant amount of attention for its ability to train highly effective deep neural networks without the need for manual annotations. SimCLR operates on the principle of contrastive learning, where the model learns to differentiate between similar and dissimilar pairs of augmented views of the same image. It aims to maximise agreement between augmented versions of the same sample while minimising agreement between samples from different classes. By doing so, it effectively learns to capture the underlying semantic information present in the data.

One of the key factors that sets it apart is its simplicity. The approach utilises a straightforward two-step training procedure. First, the model learns representations through a base

encoder network, and then a projection head is applied to map the learned representations to a latent space. By training the model with many diverse and augmented samples, SimCLR effectively learns to extract highly informative and discriminative features.

Momentum contrast (MoCo)

Unsupervised representation learning has also gained immense attention in the field of deep learning. Momentum contrast learning is one such unsupervised learning model that has proven to be a powerful framework for training highly effective neural networks without the need for manual annotations. MoCo builds upon the principles of contrastive learning, which aims to maximise agreement between similar samples